

Mechanism of Hydrogen Abstraction Reactions by Free Radicals: Simple Metathesis or Involving Intermediate Complex?

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Received: August 25, 1992; In Final Form: November 30, 1992

Hydrogen abstraction reactions, $R + HX \rightarrow RH + X$ ($R = CF_3H_{3-y}$ ($y = 0-3$), C_2H_5 ; $X = F, Cl, Br, CH_3$), have been investigated by *ab initio* methods. It is found that with reactants which are polarized with a net positive charge on the hydrogen of HX and a net negative charge on the carbon at the radical site, reactions proceed via a weakly hydrogen-bonded intermediate complex ($R \cdots H \cdots X$) prior to the formation of the transition state ($R \cdots H \cdots X$). All geometries (reactants, intermediate complexes, transition-state structures) were optimized at the (U)MP2/6-31G* level. Electron correlation energies were evaluated at the (U)MP4/6-311G**/(U)MP2/6-31G* level. For the reaction $CH_2F + HBr \rightarrow CH_3F + Br$, the Gaussian-1 (G1) theory was employed. The stability of the intermediate complexes ($R \cdots H \cdots X$) depends mainly on the dipole interaction between the polarized reactants as well as the one-electron, two-orbital interaction between the SOMO of the radical and the σ^*_{HX} orbital of HX . The theoretical results also furnish indirect support for negative activation energies found experimentally for $R + HBr \rightarrow RH + Br$ ($R = \text{alkyl radical}$). Absolute rate constants and kinetic isotope effects for the reaction $C_2H_5 + HBr(DBr) \rightarrow C_2H_6(C_2H_5D) + Br$ are evaluated by transition-state theory (TST) and RRKM theory as applied to the dissociation of the intermediate complex.

Introduction

In previous papers^{1,2} it has been shown that hydrogen abstraction reactions by CH_3 from HCl and HBr proceed via the same reaction mechanism, i.e., first via the formation of a loosely bonded intermediate complex followed by a transition-state structure. This, in itself, is contrary to what traditionally has been considered to be simple metathesis reactions. The experimental activation energies for these two reactions are remarkably different, the former ($CH_3 + HCl$)³ being positive and the latter ($CH_3 + HBr$)⁴ being negative. These findings raise a further question: do other hydrogen abstraction reactions proceed by the same mechanism as the $CH_3 + HCl$ (HBr) reactions? The fact that Seetula and Gutman,⁵ who also measured the activation energies of some halogen substituted methyl radicals with HI , observed a transition from small positive activation energies for $CH_2Cl + HI$ and $CHCl_2 + HI$ to small negative activation energies for CH_2Br and CH_2I with HI , respectively, does suggest that these reactions may have the same mechanism and that it is the relative stabilities of intermediate complexes and transition states which determine the experimentally observed activation energies. Moreover, in reactions of alkyl radicals with HBr or HI Gutman and co-workers^{4,6,7} found a trend that the activation energies become more negative with increasing complexity of the alkyl radicals in the series CH_3 , C_2H_5 , $i\text{-}C_3H_7$, $s\text{-}C_4H_9$, and $t\text{-}C_4H_9$. On the basis of these experimental results, it is reasonable to propose that hydrogen abstraction reactions ($R + HX \rightarrow RH + R$) of alkyl or halogen substituted alkyl radicals with HX ($X = F, Cl, Br, I$) proceed via the same reaction mechanism and that the molecular properties of R and X determine the sign and magnitude of the activation energy. To seek further support for this statement, several selected hydrogen abstraction reactions have been studied by *ab initio* theoretical methods and are reported here. An attempt is made to demonstrate the effect of systematic change in chemical properties of X and R on predicted activation energies. First, the series $CH_3 + HX \rightarrow CH_4 + X$ ($X = F, Cl, Br, CH_3$) was examined to reveal the effect of X substitution. Next, reactions of HBr with $CH_{3-y}F_y$ ($y = 0-3$) and C_2H_5 were chosen for the purpose of studying the effect of substitution on the radical side.

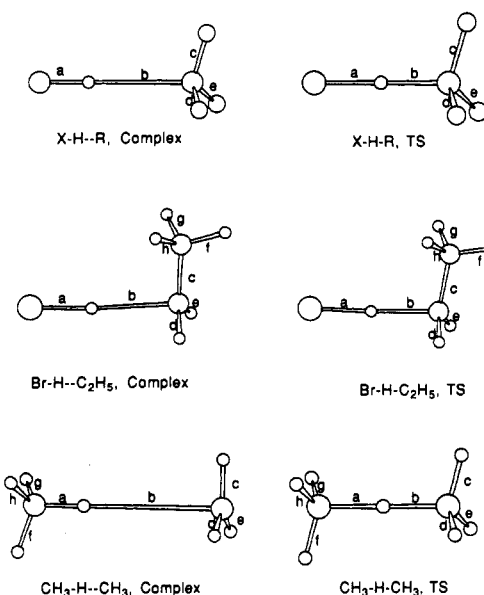


Figure 1. Geometries of the bound complex and transition-state structures for reactions $R + HX \rightarrow RH + X$ ($X = F, Cl, Br, CH_3$; $R = CH_{3-y}F_y$ ($y = 0-3$), C_2H_5). Refer to Tables I and II for numerical details.

Method

All *ab initio* calculations were carried out using the GAUSS-90 system of programs.⁸ Geometries of reactants, complexes, and transition states were first optimized at the restricted Hartree-Fock (RHF) level for closed-shell species and at the unrestricted Hartree-Fock (UHF) level for open-shell species using analytical gradients and the 6-31G* basis set.⁹ The standard 6-31G* basis set is not available for bromine. Instead, as proposed by McGrath and Radom,¹⁰ the polarized, split-valence SV4P bromine basis set of Andzelm et al.¹¹ was adopted. The 6-31G* geometries were then refined at the (U)MP2/6-31G* level¹² using all electrons for the calculation of correlation energies. Harmonic vibrational frequencies were calculated at the (U)MP2/6-31G*/(U)MP2/6-31G* levels. Electron correlation corrections were evaluated at (U)MP4/6-311G** based on the geometries optimized at the

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TABLE I: Optimized Geometrical Parameters^a for Reactants, Complexes (na), and Transition States (nb) at the (U)MP2/6-31G* Level

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>ab</i>	<i>bc</i>	<i>bd</i>	<i>be</i>	<i>cd</i>	<i>ce</i>	<i>de</i>
HF	0.934											
CH ₃		1.078	1.078	1.078						120.0	120.0	120.0
CH ₃ ...H-F, 1a	0.939	2.230	1.080	1.080	1.080	180.0	94.7	94.7	94.7	119.3	119.3	119.3
CH ₃ ...H-F, 1b	1.349	1.155	1.086	1.088	1.088	157.4	107.4	106.4	106.4	112.0	112.0	112.3
HBr	1.435											
CH ₂ F			1.350	1.082	1.082					114.3	114.3	122.0
CH ₂ F...H-Br, 2a	1.442	2.345	1.345	1.082	1.082	171.6	105.5	97.8	97.8	114.3	114.3	122.0
CH ₂ F...H-Br, 2b	1.534	1.576	1.341	1.086	1.086	178.9	106.1	100.3	100.3	113.6	113.6	119.8
CHF ₂			1.088	1.337	1.337					113.6	113.6	111.4
CHF ₂ ...H-Br, 3a	1.438	2.465	1.089	1.333	1.333	167.6	83.8	102.2	102.2	114.0	114.0	111.8
CHF ₂ ...H-Br, 3b	1.528	1.575	1.091	1.328	1.328	179.3	104.2	105.8	105.8	114.0	114.0	111.9
CF ₃			1.327	1.327	1.327					111.2	111.2	111.2
CF ₃ ...H-Br, 4a	1.435	2.655	1.325	1.325	1.325	180.0	107.4	107.4	107.4	111.5	111.5	111.5
CF ₃ ...H-Br, 4b	1.513	1.592	1.320	1.320	1.320	180.0	107.0	107.0	107.0	111.5	111.5	111.5
HCl	1.280											
CF ₃ ...H-Cl, 5a	1.281	2.721	1.325	1.325	1.325	180.0	107.4	107.4	107.4	111.5	111.5	111.5
CF ₃ ...H-Cl, 5b	1.418	1.435	1.321	1.321	1.321	180.0	107.3	107.3	107.3	111.6	111.6	111.6

^a Refer to Figure 1 for definition of parameters: e.g. *a* = bond length (Å), *ab* = bond angle (deg).**TABLE II: Optimized Geometrical Parameters^a for Reactants, Complexes (na), and Transition States (nb) at the (U)MP2/6-31G* Level**

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>ab</i>	<i>bc</i>	<i>bd</i>	<i>be</i>
C ₂ H ₅			1.489	1.082	1.082	1.100	1.093				
C ₂ H ₅ ...H·Br, 6a	1.456	2.063	1.490	1.084	1.084	1.100	1.093	175.8	98.7	95.3	95.3
C ₂ H ₅ ...H·Br, 6b	1.513	1.660	1.492	1.086	1.086	1.098	1.092	177.5	102.2	97.0	97.0
CH ₄		1.090	1.090	1.090	1.090				109.471	109.471	109.471
CH ₃ ...H·CH ₃ , 7a	1.090	3.077	1.078	1.078	1.078	1.090	1.090	180.0	90.8	90.8	90.8
CH ₃ ...H·CH ₃ , 7b	1.311	1.311	1.087	1.087	1.087	1.087	1.087	180.0	105.7	105.7	105.7
	<i>cd</i>	<i>ce</i>	<i>de</i>	<i>cf</i> (<i>af</i>) ^b	<i>cg</i> (<i>ag</i>) ^b	<i>ch</i> (<i>ah</i>) ^b	<i>fg</i>	<i>fh</i>	<i>gh</i>	<i>bcf</i> (<i>fac</i>) ^b	
C ₂ H ₅	120.7	120.7	117.2	111.9	111.4	111.4	106.9	106.9	108.0		
C ₂ H ₅ ...H·Br, 6a	119.9	119.9	116.4	110.9	111.5	111.5	107.0	107.0	108.4	180.0	
C ₂ H ₅ ...H·Br, 6b	118.9	118.9	115.3	110.2	111.5	111.5	107.5	107.5	108.6	180.0	
CH ₃ ...H·CH ₃ , 7a	120.0	120.0	120.0	109.5	109.5	109.5	109.4	109.4	109.4		
CH ₃ ...H·CH ₃ , 7b	112.9	112.9	112.9	105.7	105.7	105.7	112.9	112.9	112.9		

^a Refer to Figure 1 for definition of parameters: e.g. *a* = bond length (Å), *ab* = bond angle (deg), *abc* = dihedral angle (deg). ^b Angles in parentheses refer to structures 7a and 7b.**TABLE III: Total Energies (hartrees) for Reactants, Complexes, and Transition States at Various Levels**

	HF/6-31G*	MP2/6-31G*	MP4/6-311G** ^a	ZPVE ^b	G1 ^c
CH ₃	-39.558 992	-39.673 031	-39.730 772	19.16	-39.741 06
HF	-100.002 907	-100.184 161	-100.273 737	5.78	
CH ₃ ...H-F	-139.565 557	-139.862 599	-140.009 158	26.72	
CH ₃ ...H-F	-139.521 213	-139.814 181	-139.961 502	26.11	
HBr	-2570.722 481	-2570.851 739	-2573.107 141	3.83	-2573.164 54
CH ₂ F	-138.402 108	-138.685 954	-138.798 775	16.20	-138.890 16
CH ₂ F...H-Br	-2709.128 453	-2709.541 096	-2711.908 144	21.18	-2712.055 72
CH ₂ F...H-Br	-2709.112 070	-2709.537 306	-2711.904 380	20.98	-2712.053 72
CHF ₂	-237.263 135	-237.717 638	-237.882 501	12.43	
CHF ₂ ...H-Br	-2808.571 457	-2808.571 457	-2810.990 970	17.02	
CHF ₂ ...H-Br	-2807.969 811	-2808.566 763	-2810.987 101	16.62	
CF ₃	-336.131 184	-336.755 762	-336.970 361	7.83	
CF ₃ ...H-Br	-2907.607 978	-2907.607 978	-2910.077 692	12.14	
CF ₃ ...H-Br	-2906.834 263	-2907.601 741	-2910.073 170	11.63	
HCl	-460.059 976	-460.202 149	-460.262 778	4.36	
CF ₃ ...H-Cl	-796.958 318	-796.958 318	-797.233 271	12.75	
CF ₃ ...H-Cl	-796.164 760	-796.945 664	-797.223 516	11.58	
HBr			-2570.850 422 ^d		
C ₂ H ₅	-78.597 149	-78.844 664	-78.872 411 ^d	38.58	
C ₂ H ₅ ...H-Br	-2649.321 852	-2649.702 663	-2649.728 025 ^d	44.07	
C ₂ H ₅ ...H-Br	-2649.312 178	-2649.701 919	-2649.726 569 ^d	43.98	
CH ₄	-40.195 172	-40.337 043	-40.405 029	29.09	
CH ₃ ...H-CH ₃	-79.754 470	-80.010 894	-80.136 458	48.77	
CH ₃ ...H-CH ₃	-79.706 083	-79.974 949	-80.105 677	48.01	

^a MP4 calculations are based on geometries optimized at the MP2/6-31G* level. ^b Unscaled value (kcal/mol) at the MP2/6-31G*/6-31G* level. ^c Energy is different from G1 prescription by using ZPVE at the MP2/6-31G* level instead of the HF/6-31G* level. ^d Evaluated at the MP4/6-31G*/MP2/6-31G* level.(U)MP2/6-31G* level. For the system CH₂F + HBr → CH₃F + Br, total energy calculations were further carried out using the Gaussian-1 (G1) theory^{10,13} for the purpose of comparison withthe results reported previously^{1,2} for the reaction CH₃ + HBr → CH₄ + Br. For the G1 energy reported here, the zero-point vibrational energy (ZPVE) was evaluated at the (U)MP2/6-

TABLE IV: Calculated Energy Differences* (kcal/mol) at Various Levels

	HF/6-31G*	MP2/6-31G*	MP4/6-311G** ^b	G1
$\text{CH}_3 + \text{HF} \rightarrow \text{F} + \text{CH}_4$				
1a ^c	-0.70(-2.30)	-1.79(-3.39)	-1.32(-2.92)	
1b ^d	25.98(25.53)	27.52(27.07)	29.74(29.29)	
$\text{CH}_2\text{F} + \text{HBr} \rightarrow \text{Br} + \text{CH}_3\text{F}$				
2a ^c	-1.32(-2.42)	-1.04(-2.14)	-0.30(-1.40)	-0.66
2b ^d	8.77(7.86)	1.15(0.24)	1.87(0.96)	0.62
$\text{CHF}_2 + \text{HBr} \rightarrow \text{Br} + \text{CH}_2\text{F}_2$				
3a ^c		-0.57(-1.30)	-0.10(-0.83)	
3b ^d	10.27(9.92)	1.99(1.64)	1.94(1.59)	
$\text{CF}_3 + \text{HBr} \rightarrow \text{Br} + \text{CHF}_3$				
4a ^c		0.16(-0.30)	0.34(-0.12)	
4b ^d	12.16(12.17)	3.60(3.61)	2.71(2.72)	
$\text{CF}_3 + \text{HCl} \rightarrow \text{Cl} + \text{CHF}_3$				
5a ^c		0.29(-0.26)	0.46(-0.08)	
5b ^d	15.99(16.57)	7.11(7.69)	5.48(6.04)	
$\text{C}_2\text{H}_5 + \text{HBr} \rightarrow \text{Br} + \text{C}_2\text{H}_6$				
6a ^c	0.18(-1.39)	-2.36(-3.93)	-1.69(-3.26) ^e	
6b ^d	6.17(4.68)	-1.97(-3.46)	-0.85(-2.34) ^e	
$\text{CH}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \text{CH}_3$				
7a ^c	0.29(-0.19)	-0.03(-0.51)	0.07(-0.41)	
7b ^d	29.34(30.17)	21.81(22.04)	18.67(18.90)	
$\text{CH}_3 + \text{HCl} \rightarrow \text{CH}_4 + \text{Cl}$				
8a ^{c,f}	0.14	-1.00	-0.42	-0.67
8b ^{d,f}	11.12	4.10	4.52	2.53
$\text{CH}_3 + \text{HBr} \rightarrow \text{CH}_4 + \text{Br}$				
9a ^{c,f}	0.43	-1.04	-0.01	-0.28
9b ^{d,f}	7.41	0.38	2.11	0.67

* Energy differences include $\Delta(\text{ZPVE})$ evaluated at MP2/6-31G*//6-31G* and scaled by 0.95. Numbers in parentheses are values without $\Delta(\text{ZPVE})$. ^b For MP2/6-31G* geometry. ^c Energy difference between complex and reactants. ^d Energy difference between transition state and reactants, i.e., barrier height V_2 . ^e At the MP4/6-31G*//MP2/6-31G* level. ^f Reference 1.

31G* level instead of the (U)HF/6-31G* level as used in the original literature.^{10,13} Rate constants for the reaction of C_2H_5 with HBr and DBr were evaluated, including a tunneling correction,¹⁴ by both transition-state theory (TST) and RRKM theory, the latter applied to the dissociation of the intermediate complex as described elsewhere.²

Results and Discussions

Effect of Substituents on Intermediate Complexes and Transition States. All intermediate complexes ($\text{R}\cdots\text{H}\cdots\text{X}$) found here are linear or only slightly bent between R and HX on both the HF/6-31G* and MP2/6-31G* potential surfaces. For the reactions $\text{CHF}_2/\text{CF}_3 + \text{HBr}$ and $\text{CF}_3 + \text{HCl}$, intermediate complexes do not exist at the HF/6-31G* level but were found at the MP2/6-31G* level. Analogous to the results reported in previous papers,^{1,2} no X-coordinated complex ($\text{R}\cdots\text{X}\cdots\text{H}$), a structure postulated by Gutman and co-workers⁴ for hydrogen transfer, could be found. Structures of some of the intermediate complexes and transition states are shown in Figure 1. Geometrical parameters at the (U)MP2/6-31G* level for reactants, complexes, and transition states are listed in Tables I and II. Total energies and relative energies (i.e. energy differences) at various levels of theory are given in Tables III and IV, respectively. For purpose of comparison, Table V lists the contact C-H bond length (bond *b*) in intermediate complexes as a function of substituents. Inspection of Table V shows that intermediate complexes are only weakly bonded ($b = 2.063\text{--}3.077$ Å) and the contact distance depends on the molecular properties of X and R which are discussed below.

(a) *Effect of the Nature of X on the Contact C-H Bond Length in Intermediate Complexes.* For the series $\text{CH}_3\cdots\text{H}\cdots\text{X}$ (X = F, Cl, Br), the C-H bond (bond *b*) distance at the UMP2/6-31G* level first increases in going from F to Cl, and then decreases for

X = Br, ($b = 2.230, 2.337$, and 2.264 Å for X = F, Cl, and Br, respectively, Table V). The cause of bonding in the intermediate complex could be either the interaction of dipoles with a polarizable charge distribution and/or the orbital overlap between the SOMO of radical (CH_3) and the antibonding orbital σ^*_{HX} of HX. As the electronegativity of X decreases from F to Br (3.98, 3.16, and 2.96 for F, Cl, and Br, respectively), the HX bond becomes less polarized and the attractive interaction between CH_3 and HX would become weaker based on dipole interaction. Thus the contact bond length (bond *b*) should increase in going from F to Br which does not agree with the calculated results. However, the PMO (perturbative molecular orbital) theory does give a reasonable explanation to this trend. As shown in Figure 2, from the PMO point of view, the interaction between two approaching reactants is between the SOMO of CH_3 and the σ^*_{HX} of HX. The halogen X affects this orbital interaction in two ways: Firstly, as the electronegativity of X decreases from F to Br, the electron density of H in the σ^*_{HX} antibonding orbital becomes less, resulting in a decrease in the overlap between the SOMO and the σ^*_{HX} as X changes from F to Br. On the other hand, the energy level of σ^*_{HX} decreases as X changes from F to Br due to the change in size and energy level of the valence p orbital of X. This effect will reduce the energy gap between the SOMO and the σ^*_{HX} , resulting in an increase in the interaction and hence shortening of the bond. Therefore, the two factors cited will counteract each other, and the net of the two factors will determine the contact C-H bond length. The fact that the bond distance increases from F to Cl indicates that the electron density (overlap) factor is more important, while the energy level lowering factor dominates in the interaction when X is changed from Cl to Br.

When the CH_3 group acts as X, the resulting complex is very weak indeed. Only at the UMP2/6-31G* level is a complex found to be "stable". In fact, this complex may not be stable energetically after inclusion of the zero-point vibrational energy (ZPVE) (vide infra). The contact bond length is 3.075 Å which is larger than the C-H van der Waal's distance (2.95 Å). This is partially due to the fact that the $\sigma^*_{\text{HCH}_3}$ orbital has a much higher energy than σ^*_{HX} (X = F-Br) and is only weakly polarized, resulting in a very weak interaction between SOMO and $\sigma^*_{\text{HCH}_3}$.

(b) *Effect of Substituents in the Radical (R) on the Contact C-H Bond Length.* For the system $\text{HBr} + \text{CH}_3\cdots\text{YF}_y$ ($y = 0\text{--}3$), the contact C-H bond distance in the intermediate complexes increases as the number of fluorine substituents increases (see Table V). In PMO theory, the substitution of fluorine in the methyl radical will lower both the energy level (Table V) and the electron density of SOMO due to the high electronegativity of the F atom. Both effects will disfavor the interaction because of an increase in the energy gap between the SOMO and σ^*_{HBr} and a decrease in the overlap. This interpretation can be further verified from the result that the contact C-H bond distance in the $\text{CF}_3\cdots\text{H}\cdots\text{Cl}$ complex ($b = 2.721$ Å, Table V) is longer than that in $\text{CH}_3\cdots\text{H}\cdots\text{Cl}$ ($b = 2.337$ Å). On the other hand, when an electropositive (electron donating) group such as a methyl group is substituted for a hydrogen in the CH_3 radical, the resulting complex between CH_3CH_2 and HBr has the shortest contact length ($b = 2.063$ Å, Table V) among the complexes studied here. As expected, this is due to the fact that the methyl group raises the energy level of the SOMO orbital in CH_3CH_2 (Table V) and at the same time increases the electron density of the SOMO since the CH_3 group acts like an electron donor. However, it should be noted that the above observations of the effect of substituents in the radical (R) on the contact C-H bond length can also be rationalized on the basis of dipole interactions between R and HBr/HCl. The interaction between CF_3 and HBr or HCl is so weak that the resulting complex is not stable energetically (vide infra).

Bond Dissociation Energies, Reaction Barriers, and Vibrational Frequencies. The C-H bond dissociation energies $\text{DH}^0_0(\text{XH} -$

TABLE V: Bond Distances (Bond *b*) and Bond Dissociation Energies $DH_0^0(XH-R)^a$ in Complexes $(X-H-R)$ as a Function of Substituents (X, R)

R	SOMO ^b	bond distance (Å)				$DH_0^0(XH-R)^a$			
		X = F	X = Cl	X = Br	X = CH ₃	X = F	X = Cl	X = Br	X = CH ₃
CH ₃ CH ₂	-0.350 02			2.063				1.69 ^c	
CH ₃	-0.383 62	2.230	2.337 ^c	2.264 ^c	3.077	1.32	0.42 ^d	0.01 ^d	-0.07
CH ₂ F	-0.393 80			2.345				0.30	
CHF ₂	-0.425 17			2.465				0.10	
CF ₃	-0.478 17		2.721	2.655			-0.46	-0.34	

^a $DH_0^0(XH-R)$ refers to the energy change (kcal/mol) for the process $X-H-R \rightarrow XH + R$ at 0 K at the MP4/6-311G** level, taken from Table IV (with sign reversed corresponding to dissociation process). ^b Eigenvalue (hartree) for the single occupied molecular orbital at the MP2/6-31G* level. ^c At the MP4/6-31G**//MP2/6-31G* level. ^d Reference 1, at the MP4/6-31G**//MP2/6-31G* level.

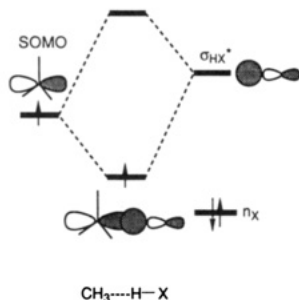


Figure 2. Perturbative molecular orbital (MPO) analysis of radical-molecule bonding showing two-orbital, one-electron interactions for a $R-H-X$ complex.

R) in intermediate complexes ($R-H-X$) which measure the enthalpy change at 0 K for the reverse process $R-H-X \rightarrow R + HX$, are listed in Table IV as "energy differences" based on the total energies and zero-point vibrational energies given in Table III. The bond dissociation energy values at the UMP4/6-31G**//UMP2/6-31G* level are selected and listed in Table V for comparison with the contact bond distances. It can be seen that the bond dissociation energies also vary with the molecular properties of X and R. The trend of the substituent effect on $DH_0^0(XH-R)$ is roughly similar to what was found for the contact C-H bond lengths except that values of $DH_0^0(XH-R)$ for CH_3-H-Br (0.01 kcal/mol) and CH_3-H-CH_3 (-0.07 kcal/mol) do not fit the trend. This may be due to the fact that ab initio methods usually contain larger errors in energy calculations (even when relative energies are considered) than in geometry optimizations and/or the fact that bond dissociation energies reflect the net stabilization of both complex and the corresponding segments (R and HX). Moreover, the relationship between stabilities and the contact distance in the complexes may be further affected by the shallowness of the attractive potential, which relates to the neglect of anharmonicity corrections. That the bond dissociation energies $DH_0^0(XH-R)$ for complexes CH_3-H-CH_3 , $Cl-H-CF_3$, and $Br-H-CF_3$ are negative indicates that these complexes are not stable energetically along the reaction coordinate, and hence such complexes may not exist.

The reaction barriers (the enthalpy difference between transition state (TS) and reactants at 0 K) at various levels of theory are also listed in Table IV. As expected, at the lower level of theory (e.g., Hartree-Fock) the computed values of reaction barriers bear little resemblance to experimental activation energies. With the inclusion of electron correlation, the disparity is considerably narrowed. Table VI lists computed reaction barriers at the MP4/6-311G** level and experimental or evaluated activation energies, E_a , for the forward reaction. Table VII lists experimental heats of formation and activation energies for the reverse processes^{3,4,15-26} used for the evaluation in Table VI. Although direct comparison cannot be made without temperature and tunneling corrections, general trends are discernable. For reactions of $CH_3 + HX$ (X = halogen), the computed reaction barriers decrease as X is changed from F to Br. This is expected since the reactions become more exothermic.

Reactions $CF_3 + HCl/HBr$ show a higher reaction barrier than the corresponding reactions with CH_3 , which is in agreement with experiment, even though these two reactions are slightly more exothermic than $CH_3 + HCl/HBr$.

For reactions of CH_2F or CHF_2 with HBr , comparison is hindered since the available radical heats of formation are not accurately known. Since at the MP4/6-311G** level reactions $HBr + CH_3-yF_y$ ($y = 1-2$) have theoretical reaction barriers similar to those of the system $HBr + CH_3$ (Tables IV and VI), it is reasonable to surmise that these reactions ($CH_2F/CHF_2 + HBr$) may also have negative temperature coefficients as found experimentally for the $HBr + CH_3$ reaction.⁴ In order to reduce electron correlation errors in the calculation of theoretical reaction barriers, the total energies were further calculated at the G1 level for the reaction $CH_2F + HBr \rightarrow CH_3F + Br$ (other systems could not be handled within the available computer disk space). The result is also listed in Table VI. It may be seen that, at the G1 level, the system $CH_2F + HBr$ shows an even smaller reaction barrier (0.62 kcal/mol) than $CH_3 + HBr$ (0.67 kcal/mol). This suggests that the rate constants for $CH_2F/CHF_2 + HBr \rightarrow CH_3F/CH_2F_2 + Br$ reactions may have a negative temperature dependence, though the results are not conclusive. Further experimental measurements for these two reactions need to be carried out to verify the theoretical prediction.

For reaction $C_2H_5 + HBr \rightarrow C_2H_6 + Br$, the transition state has a barrier of -3.46 kcal/mol; i.e., the TS has an energy lower than the reactants at the MP2/6-31G* level (Table IV). Inclusion of the zero-point vibrational energy difference, $\Delta(ZPVE)$, results in an energy difference of -1.97 kcal/mol, with the transition state still more stable than reactants (Table IV). The MP4/6-311G**//MP2/6-31G* energy calculations of both complex and transition state had computational problems (preset SCF convergence in energy could not be achieved). Hence, the electron correlation corrections were carried out at the MP4/6-31G**//MP2/6-31G* level. At this level, the transition states is 0.85 kcal/mol more stable than reactants after inclusion of $\Delta(ZPVE)$ (Table IV and VI). Earlier, Chen et al.¹ reported that the transition state for reaction $CH_3 + HBr$ is 0.38, 2.11, and 0.67 kcal/mol higher in energy than reactants at the MP2/6-31G*, MP4/6-311G**, and G1 levels, respectively.¹ Therefore, the ab initio results do support the experimental observation that the activation energies become smaller (i.e. more negative) as R changes from CH_3 to C_2H_5 (-0.31 and -0.81 kcal/mol for $CH_3 + HBr$ and $C_2H_5 + HBr$, respectively⁴).

Harmonic vibrational frequencies for reactants, intermediate complexes, and transition states were calculated at the UMP2/6-31G* level based on the geometries optimized at the same level. Tables VIII-X list the scaled ($\times 0.95$) frequencies as well as the IR intensities. The agreement between scaled harmonic frequencies and available experimental observations^{15,27-30} for reactants HF , CH_2F , CHF_2 , CF_3 , and C_2H_5 is very good at the MP2/6-31G* level (Table VIII). In the calculations of rate constants for reactions $C_2H_5 + HBr/DBr$, the theoretical frequencies have been used in anticipation that errors in frequencies for reactants and transition states will cancel out.

TABLE VI: Barrier Heights^a and Heats of Reaction^b as a Function of Substituents (X, R) in the Reaction $R + HX \rightarrow RH + X$

R	barrier ^a				$\Delta H^\circ_{298, Rx}$ ^b			
	X = F	X = Cl	X = Br	X = CH ₃	X = F	X = Cl	X = Br	X = CH ₃
CH ₃ CH ₂			-0.85 ^c (-0.81 ± 0.2)				-13.15	
CH ₃	29.74 (30.0)	4.52 ^d (1.4 ± 0.3)	2.11 ^d (-0.3 ± 0.2)	18.67 (14.1)	31.20	-1.94	-17.55	0.0
		2.53 ^e	0.67 ^e				-13.75	
CH ₂ F			1.87 (1.35 ± 2.9)				-15.65	
			0.62 ^e					
CHF ₂			1.94 (-0.07 ± 1.1)				-15.65	
CF ₃		5.48 (5.1)	2.71 (2.9)			-3.54	-19.15	

^a Barrier height refers to energy difference (kcal/mol) at 0 K evaluated at the MP4/6-311G** level between transition state and reactants after including $\Delta ZPVE$, taken from Table IV. The numbers in parentheses are experimental or evaluated activation energies based on the heat of reaction and the activation energy for the reverse reaction as given in Table VII. ^b Heat of reaction refers to enthalpy changes for reaction $R + HX \rightarrow RH + X$. The heats of formation for individual components are taken from Table VII. ^c Evaluated at the MP4/6-31G**//MP2/6-31G* level. ^d Reference 1, at the MP4/6-311G**//MP2/6-31G* level. ^e Evaluated at the G1 level.

TABLE VII: Experimental Heats of Formation for Reactants and Products and Activation Energies for the Forward Process $R + HX \rightarrow RH + X$ or Its Reverse

HX	$\Delta H^\circ_f(HX)$	ref	$\Delta H^\circ_f(X)$	ref
HF	-65.14 ± 0.2	15	18.86 ± 0.4	15
HCl	-22.063 ± 0.05	15	28.922 ± 0.06	15
HBr	-8.71 ± 0.1	15	26.74 ± 0.07	15

R	$\Delta H^\circ_f(RH)$	ref	$\Delta H^\circ_f(R)$	ref	E_f	ref	E_r	ref
X = Br								
C ₂ H ₅	-20.24 ± 0.1	16	28.36 ± 0.4	17	-0.81 ± 0.2	4		
CH ₃	-17.89 ± 0.08	15	35.1 ± 0.1	18	-0.31 ± 0.2	4		
CH ₂ F	-56.8 ± 2	19	-7.6 ± 2	20			15.10 ± 0.36	21
CHF ₂	-108.2 ± 0.2	19	-57.1 ± 1	20			15.58 ± 0.36	21
CF ₃	-166.6 ± 0.8	22	-112.0 ± 1	23	2.9	24		
X = Cl								
CH ₃					1.4 ± 0.3	3		
CF ₃					5.1	24		
X = F								
CH ₃							1.2	25
X = CH ₃								
CH ₃					14.1	26		

TABLE VIII: Vibrational Frequencies^a of Reactants at MP2/6-31G* Levels

HF			CH ₂ F			CHF ₂			CF ₃			C ₂ H ₅		
freq			freq			freq			freq			freq		
obs ^c	calc	intens ^b	obsd ^d	calc	intens ^b	obs ^c	calc	intens ^b	obs ^c	calc	intens ^b	obs ^c	calc	intens ^b
4138	3839	91.3		3207, A''	19.7		3057, A'	36.4	1260	1252, E	342.5	3112	3185, A''	13.7
				3061, A'	13.5	1317	1337, A''	88.6	1083	1070, A ₁	46.2	3033	3082, A'	12.7
				1469, A'	9.8	1173	1170, A''	191.7	701	669, A ₁	14.6	2987	3038, A''	18.0
				1153, A''	7.2	1164	1144, A'	87.1	508	479, E	2.4	2920	2995, A'	21.6
			1163	1149, A'	102.4	1053	1038, A'	13.5				2842	2914, A'	20.6
				727, A'	26.2		514, A'	5.0					1480, A''	4.7
												1440	1475, A'	3.7
													1461, A'	2.7
												1366	1397, A'	1.5
												1175	1186, A''	2.3
												1138	1054, A'	0.1
													979, A'	0.4
													799, A''	2.3
												540	441, A'	54.7
													153, A''	0.4

^a Frequencies (cm⁻¹) are calculated at the MP2/6-31G**//6-31G* level and scaled by 0.95. Values for HBr, HCl, CH₃, and CH₄ are from ref 1.

^b Intensities (kcal/mol) are calculated by the atomic polar method as implemented in GAUSSIAN 90. ^c Reference 15. ^d Reference 27. ^e Reference 28. ^f Reference 29. ^g Reference 30.

Rate Constants for Reactions $C_2H_5 + HBr/DBr \rightarrow C_2H_6/C_2H_5D + Br$. The rate constants for the reaction $C_2H_5 + HBr$ have been measured by Gutman and co-workers⁴ and more recently by Wine and co-workers.³¹ In the temperature range 295–532 K, the activation energy was reported to be -0.81⁴ and -1.07 kcal/mol.³¹ The theoretically calculated reaction barrier at 0 K is -0.85 kcal/mol at the MP4/6-31G* level which is surprisingly close to the experimental activation energies. For the purpose of further comparison, theoretical rate constants were

evaluated by both TST and the RRKM theory as described in ref 2. The results are listed in Tables XI and XII and shown in Figure 3. Since the first transition state (TS1) which separates the reactants and complex could not be found on either the HF/6-31G* or MP2/6-31G* potential surfaces, an activation barrier of $V_1 = 0.0$ kcal/mol was assumed with a separation distance of 3.5 Å between C₂H₅ and HBr, which is the same as what was done for the CH₃ + HBr system.² A reaction barrier of $V_2 = -0.85$ kcal/mol was used for the second transition state which

TABLE IX: Vibrational Frequencies^a of Complexes at MP2/6-31G* Levels

CH ₃ ··H·F		CH ₂ F··H·Br		CHF ₂ ··H·Br		CF ₃ ··H·Br		CF ₃ ··H·Cl		C ₂ H ₅ ··H·Br		CH ₃ ··H·CH ₃	
freq	intens ^b	freq	intens ^b	freq	intens ^b	freq	intens ^b	freq	intens ^b	freq	intens ^b	freq	intens ^b
3746, A ₁	321.7	3202, A''	10.2	3058, A'	27.1	2525, A ₁	32.9	2881, A ₁	48.2	3167, A''	6.2	3238, E	3.4
3224, E	0.9	3056, A'	12.2	2488, A'	96.7	1261, E	327.7	1261, A ₁	333.6	3065, A'	9.5	3089, A ₁	11.8
3046, A ₁	0.9	2433, A'	187.3	1340, A''	92.6	1069, A ₁	67.6	1070, A ₁	63.3	3047, A''	9.7	3085, E	19.8
1407, E	4.0	1467, A'	9.4	1177, A''	176.3	672, A ₁	15.5	673, A ₁	14.8	3004, A'	10.9	3059, A ₁	0.1
634, A ₁	103.7	1157, A'	111.6	1149, A'	98.8	480, E	2.4	481, E	2.4	2920, A'	9.2	2955, A ₁	0.7
340, E	122.5	1155, A''	6.4	1041, A'	13.8	123, E	18.6	143, E	42.6	2241, A'	531.6	1546, E	0.0
126, A ₁	0.5	798, A'	26.7	517, A'	4.6	32, A ₁	0.2	38, A ₁	0.2	1474, A''	7.7	1407, E	2.5
99, E	20.5	311, A''	20.9	220, A'	16.9	19, E	1.3	23, E	1.7	1473, A'	5.4	1346, E	14.4
		261, A'	22.5	197, A''	18.8					1459, A'	6.2	1341, A ₁	14.5
		126, A''	1.6	67, A'	8.6					1394, A'	4.5	405, A ₁	112.2
		79, A'	0.7	36, A'	3.9					1193, A''	2.0	88, E	0.0
		31, A'	3.7	22, A''	0.7					1050, A'	0.1	51, A ₁	0.1
										995, A'	0.4	45, E	0.0
										799, A''	2.7	4, A ₁	0.0
										650, A'	77.4		
										439, A''	15.9		
										410, A'	17.1		
										258, A''	1.5		
										99, A'	9.2		
										96, A''	0.1		
										52, A'	0.2		

^a Frequencies (cm⁻¹) are calculated at the MP2/6-31G*//6-31G* level and scaled by 0.95. ^b Intensities (kcal/mol) are calculated by the atomic polar method as implemented in GAUSSIAN 90.

TABLE X: Vibrational Frequencies^a of Transition States at MP2/6-31G* Levels

CH ₃ ··H·F	CH ₂ F··H·Br	CHF ₂ ··H·Br	CF ₃ ··H·Br	CF ₃ ··H·Cl	C ₂ H ₅ ··H·Br	C ₂ H ₅ ··D·Br	CH ₃ ··H·CH ₃
3135, A'	3164, A''	3045, A'	1279, E	1279, E	3147, A''	3147, A''	3127, E _u
3123, A''	3027, A'	1349, A''	1161, A ₁	1102, A ₁	3055, A''	3055, A''	3127, E _g
3002, A'	1464, A'	1187, A''	906, A ₁	905, E	3048, A'	3048, A'	2991, A _{1g}
1526, A'	1200, A'	1178, A'	722, E	749, A ₁	3014, A'	3014, A'	2990, A _{2u}
1458, A''	1165, A''	1139, A'	539, A ₁	482, E	2928, A'	2928, A'	1450, E _u
1409, A'	1154, A'	946, A'	482, E	334, A ₁	1479, A'	1473, A'	1434, E _g
1299, A'	862, A'	758, A''	80, E	89, E	1468, A''	1468, A''	1357, E _u
1204, A''	816, A''	716, A'	387i, A ₁	1264i, A ₁	1465, A'	1456, A'	1191, A _{2u}
947, A'	671, A'	505, A'			1455, A'	1392, A'	1171, A _{1g}
180, A'	320, A'	139, A'			1384, A'	1199, A''	699, E _g
48, A''	98, A'	83, A''			1199, A''	1125, A'	512, A _{1g}
1324i, A'	504i, A'	447i, A'			1052, A'	1051, A'	306, E _u
					996, A'	959, A'	52, A _{1u}
					803, A''	802, A''	2027i, A _{2u}
					784, A'	747, A'	
					722, A''	593, A''	
					630, A'	467, A'	
					366, A''	322, A''	
					136, A''	135, A''	
					94, A'	93, A'	
					241i, A'	236i, A'	

^a Frequencies (cm⁻¹) are calculated at the MP2/6-31G*//6-31G* level and scaled by 0.95.

TABLE XI: Calculated Rate Constants^a for Reaction C₂H₅ + HBr → C₂H₆ + Br (V₂ = -0.85 kcal mol⁻¹)

T (K)	k _{TST}	k _{RRKM}	κk _{TST} ^b	κk _{RRKM} ^b	expt ^c	expt ^d
200.0	5.24	2.57	5.89	2.89		
250.0	3.05	1.92	3.29	2.08		
295.0	2.22	1.57	2.35	1.66	2.52	
300.0	2.16	1.53	2.28	1.62		
319.0	1.95	1.43	2.05	1.50	2.27	
348.0	1.73	1.29	1.80	1.34	2.02	3.77
385.0	1.54	1.15	1.59	1.18	1.84	
400.0	1.48	1.10	1.52	1.13		
427.0	1.39	1.01	1.43	1.04	1.55	2.64
450.0	1.34	0.953	1.37	0.977		
478.0	1.29	0.885	1.32	0.904	1.51	
500.0	1.27	0.835	1.29	0.852		
532.0	1.24	0.768	1.26	0.780	1.35	

^a Units: 10¹² cm³ mol⁻¹ s⁻¹. ^b Rate constant including tunneling factor, κ ≈ 1 + (1/24)(h|ν[‡]|/k_BT)² where ν[‡] is the imaginary frequency of the transition state (TS₂), and k_B is the Boltzman constant. ^c Reference 4. ^d Reference 31.

separates complex and products. From Tables XI and Figure 3, several features can be observed. Firstly, the TST rate constants are consistently higher than those obtained from the RRKM theory, which is expected and was also observed for CH₃ + HBr,²

albeit to a much lesser degree. Secondly, the TST rate constants are slightly lower than the experimental values of Gutman and co-workers,⁴ while rate constants calculated by the RRKM theory are about 40% lower. Since Wine and co-workers³¹ reported rate

TABLE XII: Calculated Rate Constants^a for Reaction $C_2H_5 + DBr \rightarrow C_2H_5D + Br$ ($V_2 = -0.87$ kcal mol⁻¹)

<i>T</i> (K)	<i>k</i> _{TST}	<i>k</i> _{RRKM}	<i>κk</i> _{TST} ^b	<i>κk</i> _{RRKM} ^b
200.0	3.15	1.74	3.54	1.94
250.0	1.85	1.30	1.99	1.40
295.0	1.38	1.07	1.44	1.13
300.0	1.34	1.05	1.40	1.11
319.0	1.22	0.985	1.27	1.03
348.0	1.09	0.900	1.13	0.936
385.0	0.975	0.816	1.01	0.842
400.0	0.942	0.787	0.971	0.811
427.0	0.895	0.741	0.919	0.760
450.0	0.865	0.706	0.887	0.723
478.0	0.840	0.666	0.858	0.680
500.0	0.824	0.638	0.840	0.649
532.0	0.810	0.599	0.823	0.608

^a Units: 10¹² cm³ mol⁻¹ s⁻¹. ^b Rate constant including tunneling factor, *κ*.

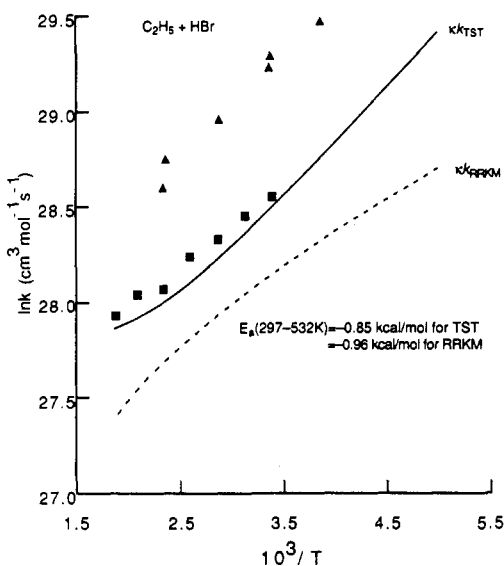


Figure 3. Arrhenius plots for $C_2H_5 + HBr \rightarrow C_2H_6 + Br$ ($V_2 = -0.85$ kcal/mol): (solid line) κk_{TST} ; (dashed line) κk_{RRKM} ; (squares) experimental data;⁴ (triangles) experimental data.³¹

coefficients for $C_2H_5 + HBr$ at least two times faster than those determined by Gutman and co-workers,⁴ the discrepancy between theoretical prediction and experiment may be even larger.

The tunneling correction evaluated by Wigner's formula¹⁴ is less important in the $C_2H_5 + HBr$ system than in the system $CH_3 + HBr$, owing to the fact that the imaginary frequency for the transition state $C_2H_5 \cdots H \cdots Br$ (241 i cm⁻¹) is less than the corresponding value for $CH_3 \cdots H \cdots Br$ (427 i cm⁻¹). The temperature dependences of TST and RRKM rate constants show a slightly different trend, the former curving upward, while the latter curving downward in the Arrhenius plots (Figure 3). A linear fit over the experimental temperature range (295–530 K) gives activation energies of -0.85 and -0.96 kcal/mol for TST and RRKM rate constants, respectively, which may be compared with the experimental values (kcal/mol) of -0.81^4 and -1.07 kcal/mol.³¹ The main difference between theoretical rate constants and experimental ones can be attributed to the pre-exponential factors (*A*). The linear fit of the theoretical Arrhenius plot over the experimental temperature range (295–530 K) (Figure 3) yields the values $A = 5.41 \times 10^{11}$ and 3.28×10^{11} cm³ mol⁻¹ s⁻¹ for the TST and RRKM rate constants, respectively, compared with the experimental values of 6.02×10^{11} ⁴ and 8.00×10^{11} cm³ mol⁻¹ s⁻¹.³¹ It should be pointed out that the RRKM calculations involve one further approximation aside from assumptions that the redistribution of internal energy is very rapid and that the first transition state is located at a contact distance of 3.5 Å. Both first and second transition states have an asymmetric rotation top ($A \neq B \neq C$) for which calculations of rotational energy

levels are quite involved and must be solved numerically.³² As an approximation, they are treated here as prolate symmetric tops ($A > B = C$) since the values of *B* and *C* are very close to each other ($A = 21.84$ and 3.88 GHz, $B = 0.92$ and 1.97 GHz, $C = 0.89$ and 1.86 HGz for first and second transition states, respectively).

Rate constants for the reaction $C_2H_5 + DBr$ are also calculated by TST and RRKM theories. The results are shown in Table XII. Although the RRKM absolute rate constant at 298 K ($k = 1.13 \times 10^{12}$ cm³ mol⁻¹ s⁻¹) is also lower than the value $k = 3.88 \times 10^{12}$ cm³ mol⁻¹ s⁻¹ determined by Wine and co-workers,³¹ the theoretical isotope effect $k(C_2H_5 + HBr)/k(C_2H_5 + DBr) = 1.47$ is in good agreement with the experimental observation (1.26²³).

Conclusion

Investigations of selected hydrogen abstraction reactions ($R + HX \rightarrow RH + X$) support the reaction mechanism found previously;^{1,2} i.e., there exists a loosely bonded intermediate complex prior to the formation of the transition state along the reaction coordinate when reactants are strongly polarized with a positive charge on the hydrogen of *HX* and a negative charge on the carbon at the radical site. Although for most hydrogen abstraction reactions with moderate TS barriers the existence of such intermediate complexes will not significantly affect the reaction kinetics (i.e., activation energy and rate constant), for reactions with low barriers (close to zero) the existence of an intermediate complex may lead to an observed negative temperature dependence. For reaction of alkyl radicals (*R*) with *HBr*, theoretical results support the experimental observation that activation energies become more negative in going from $R = CH_3$ to C_2H_5 . For the reaction $CF_3 + HBr/HCl$ stable complexes are not found and the activation energies are predicted to be positive. For $CH_2F/CHF_2 + HBr$ systems no definite conclusion can be drawn given the magnitude of the computed barriers and bond dissociation energies of the intermediate complexes within the uncertainty of theory.

The stability of an intermediate complex depends on the dipole interaction of reactants and/or the one-electron, two-orbital interaction between the SOMO of the radical and the σ^*_{HX} orbital of *HX*. In PMO, the effect of *X* on the stability of intermediates is determined by the net balance between polarization of the orbital σ^*_{HX} and its energy lowering relative to the SOMO. The substitution of electronegative atoms such as *F* for hydrogen in the radical *R* weakens the interaction between SOMO and σ^*_{HX} . On the basis of the same argument, the substitution of an electron donating group, such as CH_3 , increases the interaction.

Rate constants for reactions $C_2H_5 + HBr/DBr$ are evaluated on the basis of both TST and RRKM theory. Although the theoretical absolute rate constants are somewhat lower than the experimental results,^{4,31} the predicted kinetic isotope effect is in good agreement with the available experimental value.³¹

Acknowledgment. The financial support of the Natural Sciences and Engineering Research Council of Canada is gratefully acknowledged. Y.C. thanks the University of Calgary for a Killam Scholarship and a University Postdoctoral Fellowship. We thank Dr. A. Rauk for valuable suggestions. We also thank the Academic Computer Science of the University of Calgary for a generous grant of computer time on the Convex C120.

References and Notes

- Chen, Y.; Tschuikow-Roux, E.; Rauk, A. *J. Phys. Chem.* **1991**, *95*, 9832.
- Chen, Y.; Rauk, A.; Tschuikow-Roux, E. *J. Phys. Chem.* **1991**, *95*, 9900.

- (3) Russell, J. J.; Seetula, J. A.; Senkan, S. M.; Gutman, D. *Int. J. Chem. Kinet.* **1988**, *20*, 759.
- (4) Russell, J. J.; Seetula, J. A.; Gutman, D. *J. Am. Chem. Soc.* **1988**, *110*, 3092.
- (5) Seetula, J. A.; Gutman, D. *J. Phys. Chem.* **1991**, *95*, 3626.
- (6) Russell, J. J.; Seetula, J. A.; Timonen, R. S.; Gutman, D.; Nava, D. *F. J. Am. Chem. Soc.* **1988**, *110*, 3084.
- (7) Seetula, J. A.; Russell, J. J.; Gutman, D. *J. Am. Chem. Soc.* **1990**, *112*, 1347.
- (8) Frisch, M. J.; Head-Gordon, M.; Trucks, G. W.; Foresman, J. B.; Schlegel, H. B.; Raghavachari, K.; Robb, M. A.; Binkley, J. S.; Gonzalez, C.; DeFrees, D. J.; Fox, D. J.; Whiteside, R. A.; Seeger, R.; Melius, C. F.; Baker, J.; Martin, R. L.; Kahn, L. R.; Stewart, J. J. P.; Topiol, S.; Pople, J. A. *Gaussian 90*; Gaussian, Inc.: Pittsburgh, PA, 1990.
- (9) (a) Hehre, W. J.; Ditchfield, R.; Pople, J. A. *J. Chem. Phys.* **1972**, *56*, 2257. (b) Franci, M. M.; Pietro, W. J.; Hehre, W. J.; Binkley, J. S.; Gordon, M. S.; Defrees, D. J.; Pople, J. A. *J. Chem. Phys.* **1982**, *77*, 3654.
- (10) McGrath, M. P.; Radom, L. *J. Chem. Phys.* **1991**, *94*, 511.
- (11) Andzelm, J.; Klobukowski, M.; Radzio-Andzelm, E. *J. Comput. Chem.* **1984**, *5*, 146.
- (12) (a) Moller, C.; Plesset, M. S. *Phys. Rev.* **1934**, *46*, 618. (b) Pople, J. A.; Krishnan, R.; Schlegel, H. B.; Binkley, J. S. *Int. J. Quantum Chem.* **1978**, *14*, 545. (c) Krishnan, R.; Frisch, M. J.; Pople, J. A. *J. Phys. Chem.* **1980**, *72*, 4244. (d) Krishnan, R.; Binkley, J. S.; Pople, J. A. *J. Chem. Phys.* **1980**, *72*, 650.
- (13) (a) Pople, J. A.; Head-Gordon, M.; Fox, D. J.; Raghavachari, K.; Curtiss, L. A. *J. Chem. Phys.* **1989**, *90*, 5622. (b) Curtiss, L. A.; Jones, C.; Trucks, G. W.; Raghavachari, K.; Pople, J. A. *J. Chem. Phys.* **1990**, *93*, 2537.
- (14) (a) Wigner, E. P. *Z. Phys. Chem., Abt. B* **1932**, *19*, 203. (b) Johnston, H. S. *Gas-Phase Reaction Rate Theory*; Ronald Press: New York, 1966; pp 133-136.
- (15) *JANAF Thermochemical Tables*, 2nd ed., Stull, D. R., Prophet, H., Eds.; National Standard Reference Data Series No. 37; National Bureau of Standards, U.S. Government Printing Office: Washington, DC, 1971.
- (16) Chao, J.; Wilhoit, R. C.; Zwolinski, B. J. *J. Phys. Chem. Ref. Data* **1973**, *2*, 427.
- (17) Brouard, M.; Lightfoot, P. D.; Pilling, M. J. *J. Phys. Chem.* **1986**, *90*, 445.
- (18) Heneghan, S. P.; Knoet, P. A.; Benson, S. W. *Int. J. Chem. Kinet.* **1981**, *13*, 677.
- (19) Rodgers, A. S.; Chao, J.; Wilhoit, R. C.; Zwolinski, B. J. *J. Phys. Chem. Ref. Data* **1974**, *3*, 117.
- (20) Pickard, J. M.; Rodgers, A. S. *Int. J. Chem. Kinet.* **1983**, *15*, 569.
- (21) Miyokawa, K.; Tschuikow-Roux, E. *J. Phys. Chem.* **1992**, *96*, 7328.
- (22) Chen, S. S.; Wilhoit, R. C.; Zwolinski, B. J. *J. Phys. Chem. Ref. Data* **1976**, *5*, 571.
- (23) Tschuikow-Roux, E.; Paddison, S. *Int. J. Chem. Kinet.* **1987**, *19*, 15.
- (24) Amphlett, J. C.; Whittle, E. *Trans. Faraday Soc.* **1966**, *62*, 1622.
- (25) Fettis, G. C.; Knox, J. H.; Trotman-Dickenson, A. F. *J. Chem. Soc.* **1960**, 1064.
- (26) Creak, G. A.; Dainton, F. S.; Ivin, K. J. *Trans. Faraday Soc.* **1962**, *58*, 326.
- (27) Endo, Y.; Yamada, C.; Saito, S.; Hirota, E. *J. Chem. Phys.* **1983**, *79*, 1605.
- (28) Jacox, M. E. *J. Mol. Spectrosc.* **1980**, *81*, 349.
- (29) Snelson, A. *High Temp. Sci.* **1970**, *2*, 70.
- (30) Pacansky, J.; Schrader, B. J. *J. Chem. Phys.* **1983**, *78*, 1033.
- (31) Nicovich, J. M.; van Dijk, C. A.; Kreutter, K. D.; Wine, P. H. *J. Phys. Chem.* **1991**, *95*, 9890.
- (32) Herzberg, G. *Molecular Spectra and Molecular Structure. II. Infrared and Raman Spectra of Polyatomic Molecules*; Van Nostrand Co.: Princeton, NJ, 1945.